Gunjan Barua, Ph.D.

Technical Product Management | Digital MRV · Carbon & Environmental Data Verification · Earth Observation & AI
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U.S. permanent resident; no visa sponsorship required.

Summary

Technical product specialist in digital MRV, building environmental data that stands up to audit. Experience on all three sides of carbon measurement: developed the measurement models (Ph.D. research applying AI to LiDAR and SAR for forest yield and biomass), built an end-to-end digital MRV system at a startup (CarboBon), and now turn carbon methodologies into software inside the standard-setter (Verra). Translate standards and science into product requirements, APIs, and tested releases. Grounded in nature-based carbon markets, extending into supply-chain (Scope 3) emissions and broader environmental intelligence.

Core Competencies

- **Product & platforms:** product requirements, API design and documentation (OpenAPI/Swagger, Postman), QA/UAT and release management, partner onboarding, cross-functional coordination, workshop facilitation
- **Carbon & GHG accounting:** VCS Program (incl. v5.0), AFOLU methodologies, digital MRV, GHG quantification (Scope 1–3), Guardian (Hedera)
- **Earth observation & AI:** UAV and airborne LiDAR, C-band SAR, multispectral/hyperspectral, Google Earth Engine, Random Forest, SVR, XGBoost, LightGBM, LSTM/GRU
- **Generative AI & automation:** AI workflow and agent design (Claude, Claude Code, Microsoft 365 Copilot, Gemini), prompt and context engineering, AI-assisted calculation debugging and API payload validation, LLM-based document extraction, research synthesis, human-in-the-loop verification of AI output
- **Tools:** Python, R, SQL, scikit-learn, TensorFlow, GeoPandas, GDAL, ArcGIS Pro, QGIS, Git, Streamlit

Professional Experience

Verra — Digital MRV Solutions Specialist, Technology Solutions

Remote · June 2026 – Present

- **Data-provider program:** Built and led the end-to-end onboarding pipeline for remote-sensing data service providers (DSPs) under VM0047, with 12+ providers in onboarding and integration. Run every stage for each: administrative onboarding with Legal, Billing and Communications, API integration, testing and technical troubleshooting, and mapping provider biomass outputs to methodology inputs.
- **Methodology digitalization:** Took six AFOLU policies (VM0042, VM0045, VM0047, VM0051, VM0033, AR-ACM0003) to release on Verra's Guardian (Hedera) platform, with daily syncs with Hashgraph. On Project Hub, scoped & tested methodology updates (VM0042, VMD0055, VM0033, VM0051) through to production.
- **Credit integrity:** Diagnosed and corrected calculation-engine issues across methodologies affecting credit outcomes (removals vs. reductions, platform-wide memory issue) and shipped the fixes to production.
- **Standards to software:** Translated VCS v5.0 templates into developer requirements and designed the calculation-engine flow for grouped projects, validated with a full-chain example. Contributed to the template for Verra's new Scope 3 standard and tested its project description in QA.
- **Product & documentation:** Wrote requirements for the public vetted-DSP directory (live on Project Hub); authored API documentation for three methodologies, a reusable template, and a "Getting Started" guide; set up Swagger/Postman testing.
- **Innovation:** Facilitated a cross-team roundtable on Verra's digital carbon ecosystem; built an AI agent that tracks external dMRV and AFOLU developments weekly.

CarboBon Inc. — Technical Advisor

Remote · May 2023 – October 2025

- Designed and deployed an end-to-end digital MRV system using satellite imagery and machine learning for near-real-time forest monitoring and GHG quantification.

- Defined data schemas and automated validation logic for carbon project datasets, improving data integrity and audit readiness.
- Translated carbon methodology requirements into digital workflows; built an API for corporate Scope 1 and 2 emissions.

Virginia Tech — Graduate Research Assistant, Forest Resources & Environmental Conservation

Blacksburg, VA · January 2023 – May 2026

- Developed ML models that estimate loblolly pine yield and biomass from UAV LiDAR and Sentinel-1/2 data (±2% stand-level bias; $R^2 = 0.90$ in a multi-year transferability test); published in *Forest Ecology and Management* and the *International Journal of Remote Sensing*.
- Engineered reproducible pipelines integrating LiDAR, SAR and optical imagery; packaged the dissertation as an interactive dashboard.
- With the Forest Productivity Cooperative and Michigan State University, built an LLM-assisted pipeline that digitizes handwritten field records and receipts, cutting manual processing time by 70%.

Earlier Experience

- **Graduate Researcher & Instructor**, Virginia Tech (2021–2023): developed a UAV thermal-mapping method validated in a 66-participant study; taught Advanced Remote Sensing to 232 students.
- **GIS Developer Intern**, GFR Forestry Consultants (2023): rebuilt ArcGIS Online dashboards and automated field workflows with Survey123, Power Automate and the ArcGIS API for Python.
- **Policy Advocacy Specialist & Documentation Officer**, Municipal Association of Bangladesh and Practical Action (2016–2020): managed $850K in program portfolios and co-drafted a national WASH action plan.

Education

- **Ph.D., Geospatial and Environmental Analysis**, Virginia Tech, 2026
- **M.S., Geography**, Virginia Tech, 2023; **Graduate Certificate, Geospatial Information Technology**, 2022
- **Bachelor of Urban and Regional Planning**, Khulna University of Engineering & Technology, Bangladesh, 2016

Selected Publications

10 peer-reviewed papers and manuscripts in remote sensing, AI, and environmental risk; full list on the website.

- **Barua, G.**, et al. (2026). Estimating *Pinus taeda* timber volume with multi-temporal SAR–optical fusion and supervised learning. *International Journal of Remote Sensing*.
- **Barua, G.**, Pires, M., & Kim, J. (2026). Static maps, dynamic threats: Re-evaluating U.S. wildfire risk with spatiotemporal clustering. *Natural Hazards*.
- Platt, E., … **Barua, G.**, & Sumnall, M. J. (2026). Precision forestry: Using AI and LiDAR to inform thinning in *Pinus taeda* plantations. *Journal of Forestry*.
- **Barua, G.**, et al. (2025). Predicting yield of *Pinus taeda* using LiDAR data in random forest and support vector machine models. *Forest Ecology and Management*.
- Islam, K. M. A., **Barua, G.**, et al. Transfer learning with pretrained AlphaEarth embeddings for mangrove forest mapping in Bangladesh. *Discover Forests* (under review).

Selected talks: VM0047 onboarding webinar for project developers (co-host with Hashgraph), Verra, 2026 · AGU 2025 · ASPRS at Geo Week 2025 · Weyerhaeuser (invited), 2025

Awards, Certifications & Service

- **Awards:** 2nd Prize, OGIS Symposium Poster (2026) · Best Student Presentation, APSAF (2025) · Sigma Xi Research Award (2022) · 1st Prize, OGIS Graduate Poster (2022) · Georgia-Pacific Fellowship (2021)
- **Certifications:** FAA Part 107 Remote Pilot
- **Peer reviewer:** *Scientific Reports*, *Cartography and Geographic Information Science*, *Remote Sensing Applications: Society and Environment*, and eight other journals
- **Affiliations:** ASPRS · American Geophysical Union · Society of American Foresters · Sigma Xi · CaGIS